Lab 2: Software Requirements Specification

Miguel Looney

CS 411

Thomas J. Kennedy

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and nonfunctional requirements for the Gold Star prototype. This document describes the expected capabilities, user interactions, and overall characteristics of the system. It provides a reference for the development, evaluation, and testing of Gold Star.

1.2 Scope

Attention-Deficit/Hyperactivity Disorder (ADHD) affects many adults, and can make it difficult to build routines, stay organized, and manage time (CDC, 2024). These challenges can make everyday responsibilities harder to complete and create the need for tools that help users stay on track. Gold Star was created to provide a simple and flexible solution to these problems.

Gold Star is a web-based habit-building and time management platform designed for individuals with ADHD and ADHD-like traits. It helps users organize tasks, build routines, and manage their schedules through an adaptive system that adjusts when tasks are missed or priorities change. This prototype demonstrates the application's main features and how they can support productivity and long-term routine building.

1.3 Definitions, Acronyms, and Abbreviations

- Activity Alternation – Switching from an activity requiring high mental focus to an activity requiring low mental focus.
- Activity Switching – The cognitive process of disengaging from one task and shifting mental focus to another task.
- Adaptive Scheduling – To schedule a task which can be completed anytime within a set timeframe. Example: a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

- Application Programming Interface (API) - A set of standard rules, protocols, and definitions allowing different software applications to interact with one another.
- Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.
- Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.
- Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.
- GitHub - A cloud-based platform where developers can collaborate on software code.
- Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.
- Habit Stacking – Interspersing new habits with existing habits to make task completion easier without feeling overwhelmed.
- Hyperactivity - A state of being unusually or abnormally active.
- HyperText Markup Language (HTML) - The standard markup language used to create and structure webpages.
- JavaScript (JS) - A lightweight programming language used to build dynamic interactivity on web pages.
- Jest - An open-source JavaScript testing framework.
- JSDoc - An API documentation generator and syntax standard for JavaScript.

- Non-Rigid Task – Tasks lacking required starting times and deadlines. Non-Rigid Tasks require an estimated timeframe for completion. Non-Rigid Tasks are scheduled by Adaptive Scheduling.
- Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats.
- PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.
- Reinforce - To make something stronger, more effective, or more compelling by adding support.
- Relational Database Management System (RBMS) - Software used to store, manage, query, and retrieve data structured in table form.
- Rigid Scheduling – To schedule a high-priority task which must either be initiated at or completed before an established time. Example: a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.
- Rigid Task – Non-negotiable tasks requiring a specific starting time or deadline for completion. Rigid Tasks are scheduled by Rigid Scheduling.
- Routine Adaption – A requirement for an individual to change one's schedule or create new routines.
- Routine - A sequence of actions regularly followed, or a fixed program.
- Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

- Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

1.4 References

Team Gold Star. (2026, September 28). *Lab 1 – Gold Star Product Description.* Old Dominion University.

1.5 Overview

Section 2 provides a generalized description of Gold Star, including its product perspective, functions, users, constraints, and dependencies. Section 3 of the complete SRS specifies the functional and nonfunctional requirements for the planned prototype.

2 Overall Description

2.1 Product Perspective

Gold Star is a web-based productivity and habit-building platform designed primarily for individuals with ADHD who struggle with routine creation and maintenance. The application provides a low-stakes environment in which users can organize tasks, view schedules, and respond to changes in their daily plans.

Gold Star is accessible through a web browser without specialized hardware. The prototype uses a web interface developed with HTML, CSS, and JavaScript, a Node.js/Express server, and a MySQL database for task-related information.

2.2 Product Functions

The planned prototype provides the following core task-management and routine-building functions:

- Task Management - Users can create, edit, delete, and schedule tasks.
- Task Prioritization - Users can assign tasks a High, Medium, or Low priority.

- Calendar View - Users can view scheduled tasks by date.
- Adaptive Scheduling - Users can adjust or reschedule tasks when their plans change.
- Habit Stacking - Users can associate tasks to support routines.
- Daily Check-In - Users can record their mood or productivity level.
- Account Management - Users can create and access an account, if included in the agreed prototype scope.

2.3 User Characteristics

The primary user of Gold Star is the Scheduler, typically an adult with ADHD or ADHD-like difficulties. Schedulers manage tasks, organize calendars, assign priorities, and build routines. The application is intended to accommodate users with varying levels of technical experience.

Other people who struggle with organization, productivity, or habit-building consistency may also benefit from the application. The development team uses the SRS to implement and evaluate the prototype but is not a separate end-user role.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A